End of ertainty

Experience is the best teacher, but not in technology, until recently. Find out how a 250-year-old theorem is being employed in today's systems to make use of the knowledge of the past

never a surety, but more a recommendation. For example, a Bayesian learner might say, after observing the weather, that it'll be sunny tomorrow, with an 80 per cent probability. Though it might well rain the next day, the 80 per cent probability is a certainty. It's up to us to decide how to react to it. Some of us might disbelieve it, others may plan a trip to the beach, while still more might toss a coin thrice, and only believe the 'sunny' outcome if the coin turns up tails twice or more.

So, a Bayesian learner can predict outcomes based on data. But so can Neural Networks (NNs), and various other AI models. What's so special about Bayesian? To begin with, a Bayesian learner can make predictions incrementally, that is, as more knowledge is fed into the system, the system's prediction changes instantaneously. Secondly, BBNs are not 'black boxes' like NNs, where no one can know what's going on in the system. Also, NNs generally give yes or no answers, while a probabilistic estimate is often preferable. With BBNs, the 'answer' is updated as soon as data is updated, which is not the case with NNs. BBNs are more robust than rule-based methods and other statistical methods. *(See box 'Method Madness')*

Where have I seen a BBN?

Wherever anything needs to be predicted, and wherever a pattern is to be found, BBNs can work. Both these—prediction and pattern finding—can come from learning from the past; when you observe something over a period of time, you can predict what's going to happen, and find patterns in what you've observed.

Bayesian methods you see every day include spam filters and also the cute but immensely irritating assistants in MS Office. The Office assistants learn from the mouse clicks you make, the text you enter in your searches, and such. Printer troubleshooters, too, benefit from Bayesian knowledge.

Have you noticed how personalised content delivery on the Net seems to

Method Madness

Here's a comparison of three learning methods: rule-based methods (make and find rules and stick to them), Neural Nets, and BBNs. An example application is spam filtering. Suppose a rule has been learnt by all three systems, a rule that says that any e-mail with the word 'debt' in it is spam. And now, suppose you work for a debt-consulting agency for just three days, so all legitimate e-mails contain the word 'debt' in them. You want to classify these mails as 'not spam'. What would happen in each case?

Rule-based learning	Neural Nets	BBNs
You would have to change the rule manually, and after the three days, the system would need to relearn the rule from scratch.	The system gives a yes or no answer about spam, and it would take a lot of learning—many e-mails—for it to learn that your new job mails aren't spam.	The system only gives probabilities. And, after you get a few e-mails with the word 'debt' in them, and you say they aren't spam, the system would look for other factors in those e-mails that make them non-spam.